

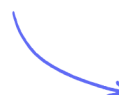
IGCSE · Cambridge (CIE) · Maths

 25 mins  6 questions

Investigation: Ellipses

Investigations

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Total Marks

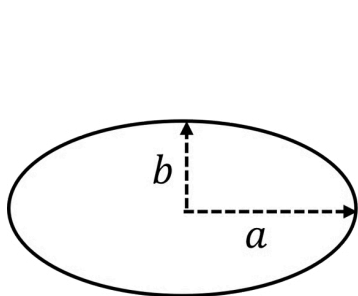
/25

- 1 (a)** This investigation looks at the geometry and properties of an ellipse, including its area and its circumference.

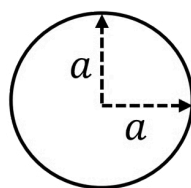
An ellipse can be thought of as a squashed or stretched circle.

It can be described by a horizontal length, a , and a vertical length, b , measured from its centre, as shown below.

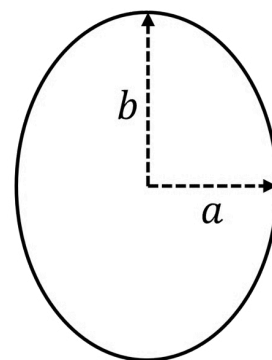
When $a = b$, the ellipse is a circle with radius a .



$$a > b$$



$$a = b$$



$$a < b$$

This investigation will only consider ellipses in the form $a \geq b$.

Sketch an ellipse with $a = 5$ and $b = 2$.

Label the two lengths on your diagram.

(2 marks)

- (b)** The formula for the area of an ellipse, A , is given by

$$A = \pi ab$$

Complete the table below, leaving all answers in terms of π .

The first example has been done for you:

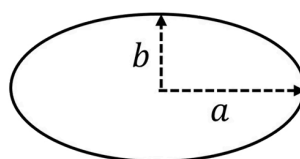
$$A = \pi \times 5 \times 2 = 10\pi$$

a	b	A
5	2	10π
8	3	
	3	36π
10		50π

(1 mark)

- 2 An ellipse is a shape that can be thought of as a squashed or stretched circle.

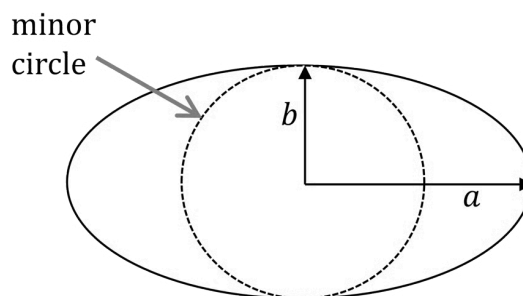
It can be described by a horizontal length, a , and a vertical length, b , measured from its centre, as shown below.



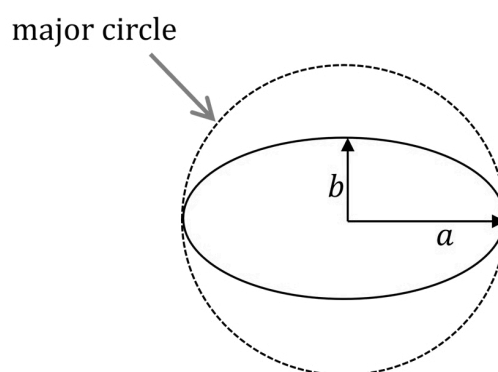
The formula for the area of an ellipse, A , is given by

$$A = \pi ab$$

A minor circle refers to the biggest circle that fits **inside** an ellipse, with the same centre, as shown.



A major circle refers to smallest circle that fits **outside** an ellipse, with the same centre, as shown.



The following notation will be used for the different areas:

- A_E is the area of an ellipse
- A_C is the area of its minor circle
- A_D is the area of its major circle

Complete the table below, leaving answers fully simplified and in terms of π (where necessary).

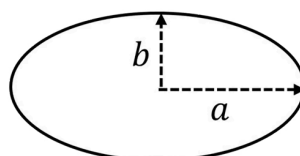
Parts of the table have been done for you.

a	b	$\frac{a}{b}$	A_E	A_C	$\frac{A_E}{A_C}$	A_D	$\frac{A_E}{A_D}$
8	2	4	16π	4π	4	64π	$\frac{1}{4}$
10	5	2	50π	25π	2		
12	4						

(3 marks)

- 3 An ellipse is a shape that can be thought of as a squashed or stretched circle.

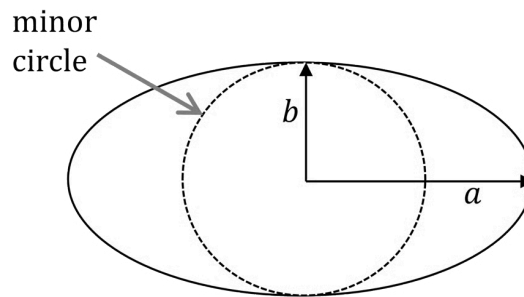
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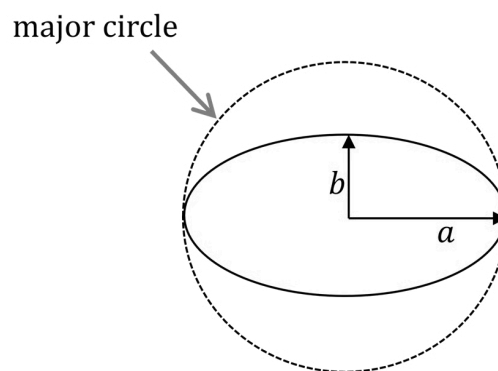
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The following notation is used for the different areas:

- A_E is the area of an ellipse
- A_C is the area of its minor circle
- A_D is the area of its major circle

Use the area formulas for A_E , A_C and A_D to prove **algebraically** that

$$\frac{A_E}{A_C}$$

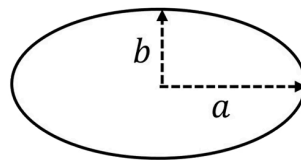
is the reciprocal of

$$\frac{A_E}{A_D}$$

(3 marks)

4 (a) An ellipse is a shape that can be thought of as a squashed or stretched circle.

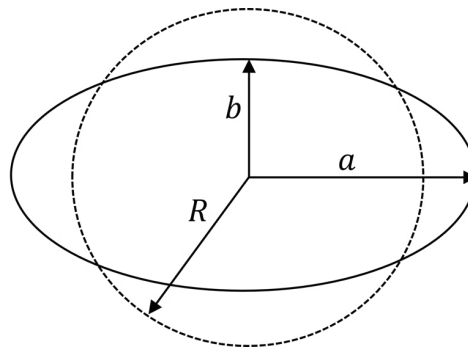
It can be described by a horizontal length, a , and a vertical length, b , measured from its centre, as shown below.



The formula for the area of an ellipse, A , is given by

$$A = \pi ab$$

In the diagram below, the area of the circle, radius R , is equal to the area of the ellipse, πab .



A circle, radius R , with the same area as the ellipse

Use algebra to show that

$$R = \sqrt{ab}$$

(2 marks)

(b) There is no algebraic formula for the **circumference** of an ellipse, C .

However, there are many formulas that give approximations to the circumference of an ellipse in the form

$$2\pi r$$

One approximate formula uses $r = \sqrt{ab}$ from part (a), giving:

$$C_1 = 2\pi\sqrt{ab}$$

A second approximate formula uses r as the mean of a and b , giving:

$$C_2 = 2\pi\left(\frac{a+b}{2}\right)$$

The table below compares the approximations C_1 and C_2 to the true circumferences, C .

Complete the table, giving answers correct to 1 decimal place.

The first two rows have been done for you.

a	b	C	C_1	C_2
10	9	59.7	59.6	59.7
10	1	40.6	19.9	34.6
15	13	88.1		
15	2	61.6		

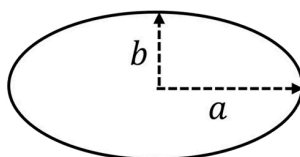
(2 marks)

- (c) Explain, with evidence from the table in part (b), which approximation out of C_1 and C_2 you would **not** recommend for ellipses that are very flat.

(2 marks)

5 (a) An ellipse is a shape that can be thought of as a squashed or stretched circle.

It can be described by a horizontal length, a , and a vertical length, b , measured from its centre, as shown below.

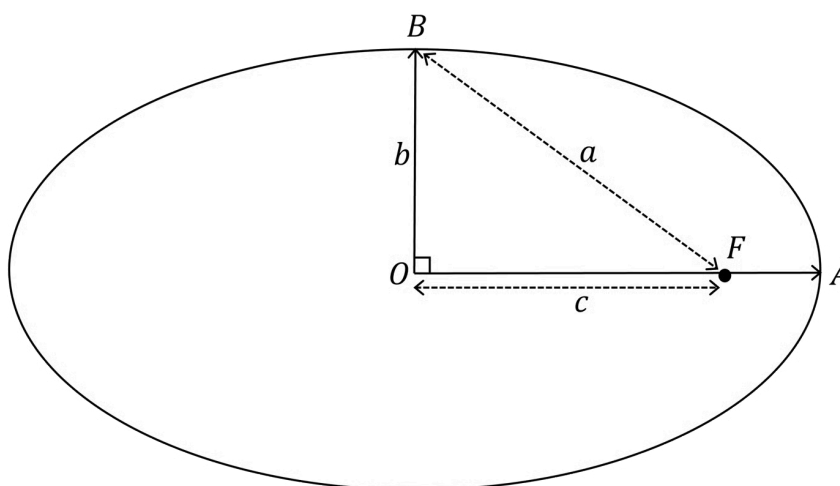


The point F on the diagram below is called the focus of an ellipse.

The focus is:

- a point on the line OA
- that is a distance c away from O
- and that forms a right-angled triangle, BOF
- with a hypotenuse of length a , as shown.

The length $OA = a$ is not shown.



The eccentricity of an ellipse, e , is the ratio of the length c to the length a , given by

$$e = \frac{c}{a}$$

By finding c in terms of a and b from the diagram, show that the eccentricity can be written in the form

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

(1 mark)

- (b) Two formulas for approximating the circumference of an ellipse are $C_1 = 2\pi\sqrt{ab}$ and $C_2 = 2\pi\left(\frac{a+b}{2}\right)$.

A third formula to approximate the circumference of an ellipse is

$$C_3 = 2\pi a \left(1 - \frac{1}{4}e^2 - \frac{3}{64}e^4\right)$$

where e is the eccentricity of the ellipse, as given in part (a).

Use this formula for C_3 to work out an estimate for the circumference of an ellipse with $a = 26$ and $b = 24$.

You must show your working clearly.

Give your answer correct to 2 decimal places.

(3 marks)

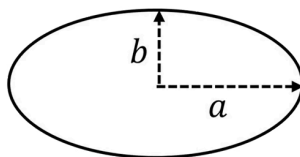
- (c) The true circumference of an ellipse with $a = 26$ and $b = 24$ is 157.14, rounded to 2 decimal places.

Explain whether the estimate in part (b) is an overestimate or an underestimate.

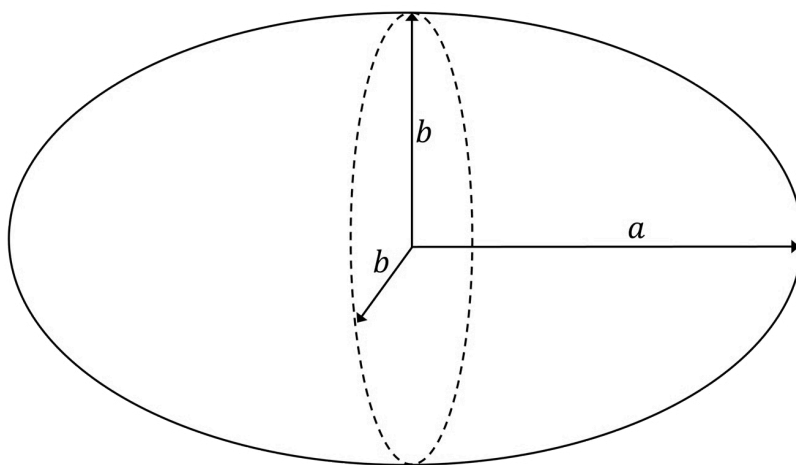
(1 mark)

- 6 An ellipse is a shape that can be thought of as a squashed or stretched circle.

It can be described by a horizontal length, a , and a vertical length, b , measured from its centre, as shown below.



An ellipse can be rotated 360° about the x -axis to form a 3D shape called a prolate spheroid, as shown, where $a > b$.



The total surface area of a prolate spheroid, A , is given by the formula

$$A = 2\pi a^2 \left[1 - e^2 + \frac{\sqrt{1 - e^2}}{e} \left(\frac{\theta\pi}{180} \right) \right]$$

where $e = \frac{\sqrt{a^2 - b^2}}{a}$ is the eccentricity of the ellipse, and θ is the acute angle in degrees that satisfies the equation

$$\sin \theta = e$$

Use the information above to find the total surface area of a prolate spheroid with $a = 2$ and $b = \sqrt{3}$.

Leave your answer in the exact form

$$A = p\pi + q\sqrt{3}\pi^2$$

where p is an integer and q is a fraction, both of which you should find.

(5 marks)